Exp No : Date:	SMART FIREFIGHTER IoT SYSTEM USING ARDUINO – MINI PROJECT

OBJECTIVE:

To design and simulate a Smart Firefighter IoT System using an Arduino Uno and two TMP36 temperature sensors. The system monitors temperature in two locations and switches ON a water-pump-representing motor, buzzer, and red LED when either sensor reaches the specified threshold. A green LED indicates the safe condition.

COMPONENTS REQUIRED:

S.NO	COMPONENTS	QUANTITY
1.	Arduino Uno R3	1
2.	TMP36 Temperature Sensor	2
3.	DC Motor (water pump representation)	1
4.	NPN Transistor	1
5.	Buzzer	1
6.	Flyback Diode	1
7.	Connecting wires and breadboard	As required
8.	Tinkercad Circuits simulation software	1

PROCEDURE:

STEP 1 : Open Tinkercad Circuits and create a new circuit.

STEP 2 : Place the Arduino Uno, two TMP36 sensors, DC motor, NPN transistor, buzzer, red LED, green LED, resistors, and connecting wires.

STEP 3 : Connect the first TMP36 sensor output to A0 and the second TMP36 sensor output to A1. Connect their power and ground pins to 5V and GND.

STEP 4 : Connect the motor control circuit to digital pin 5 through the NPN transistor driver circuit.

STEP 5 : Connect the buzzer positive terminal to digital pin 8 and its negative terminal to GND.

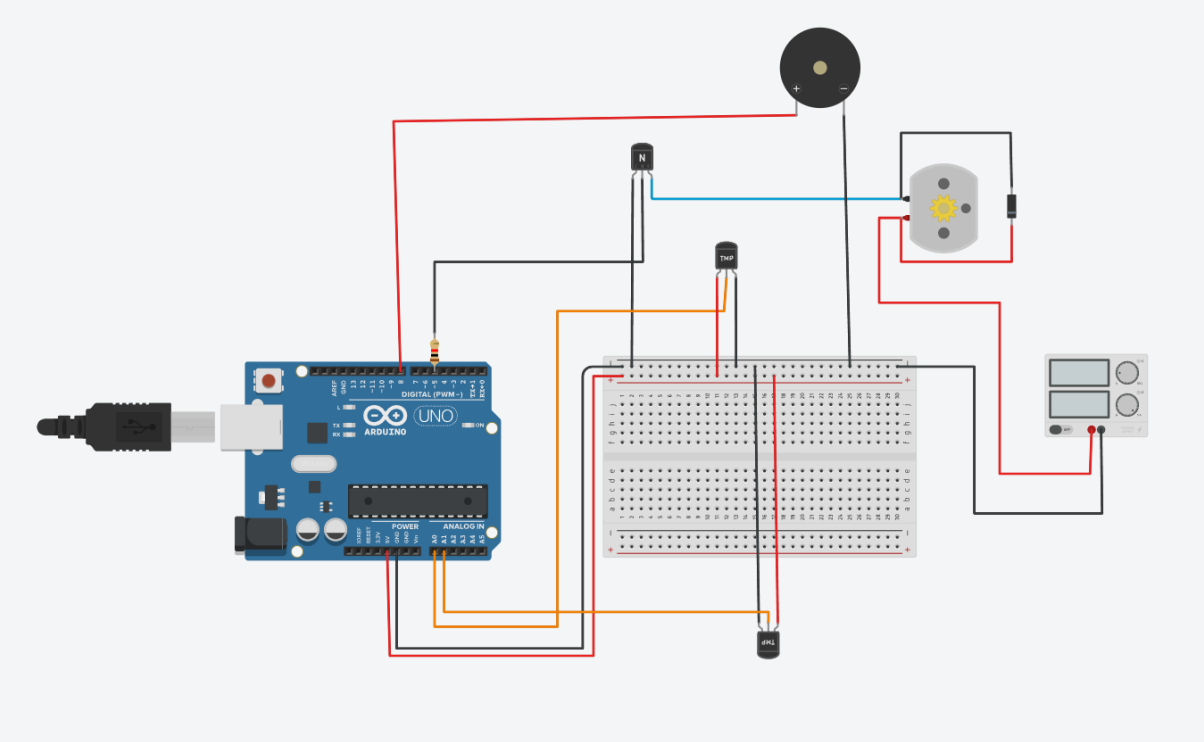
STEP 6 : Connect the green LED to digital pin 9 and the red LED to digital pin 10, each through a current-limiting resistor.

STEP 7 : Write the Arduino program in the Tinkercad code editor using Text mode.

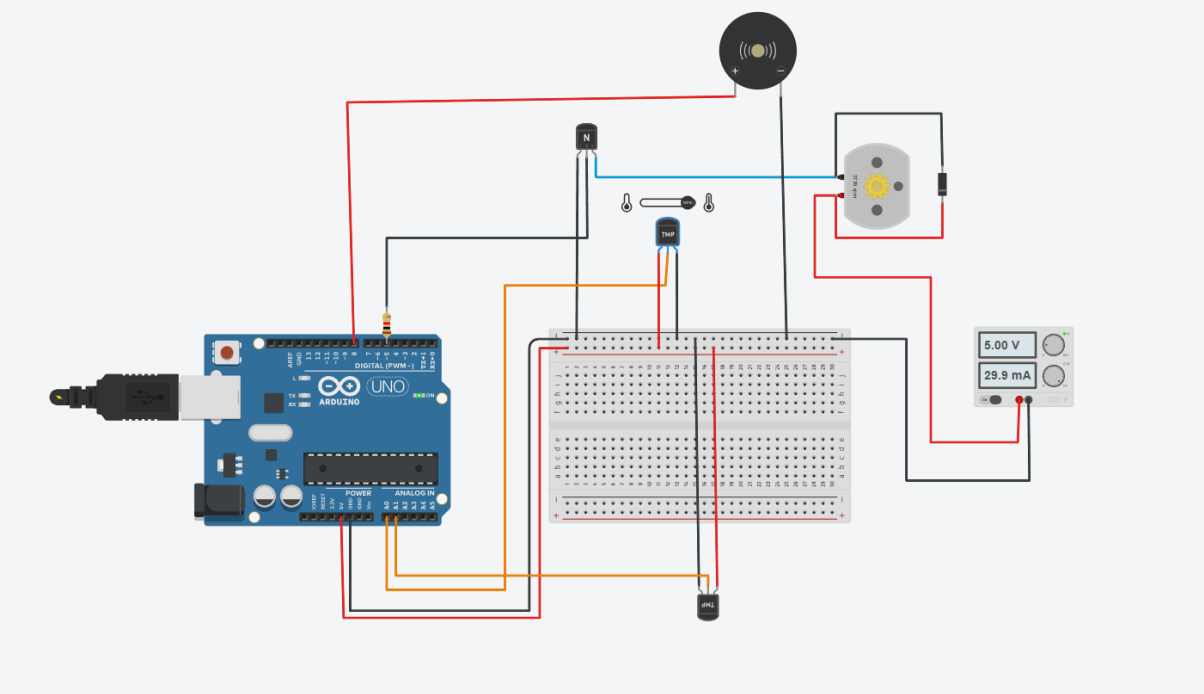
STEP 8 : Start the simulation and vary the temperature settings of the TMP36 sensors.

STEP 9 : Observe the pump, buzzer, LED indications, and temperature readings in the Serial Monitor.

OUTPUT:(BEFORE SIMULATION)



OUTPUT:(AFTER SIMULATION)



COMPONENTS DEFINITION:

1. Arduino Uno R3: A microcontroller board that reads the temperature sensors and controls the motor driver, buzzer, and LEDs according to the program.

2. TMP36 Temperature Sensor: Measures temperature and provides an analog voltage to the Arduino. Sensor 1 is connected to A0 and sensor 2 is connected to A1.

3. DC Motor: Represents the water pump. It turns ON when the system detects a high-temperature condition and turns OFF in the safe condition.

4. NPN Transistor: Acts as an electronic switch to control the motor because the Arduino output pin cannot safely supply the motor current directly.

5. Buzzer: Produces an audible alert when either temperature sensor reaches the programmed threshold.

6. Red LED: Indicates the high-temperature or fire-alert condition.

7. Green LED: Indicates that both temperature sensors are below the threshold.

8. Resistors: Limit current through the LEDs and provide appropriate base current to the NPN transistor.

9. Flyback Diode: Provides a path for inductive current and helps protect the transistor from voltage spikes causethe motor.

10. Connecting Wires and Breadboard: Provide temporary electrical connections between the Arduino and the components.

PROGRAM:

```
const int tempPin1 = A0;
const int tempPin2 = A1;
const int pumpPin = 5;
const int buzzerPin = 8;
const int redLED = 10;
const int greenLED = 9;
const float fireThreshold = 50.0;

float readTemperature(int pin) {
  int reading = analogRead(pin);
  float voltage = reading * (5.0 / 1023.0);
  return (voltage - 0.5) * 100.0;
}

void setup() {
  pinMode(pumpPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  float temp1 = readTemperature(tempPin1);
  float temp2 = readTemperature(tempPin2);

  Serial.print("Sensor 1 (A0): ");
  Serial.print(temp1);
  Serial.println(" C");

  Serial.print("Sensor 2 (A1): ");
  Serial.print(temp2);
  Serial.println(" C");
```

```
  if (temp1 >= fireThreshold || temp2 >= fireThreshold) {
    digitalWrite(pumpPin, HIGH);
    digitalWrite(redLED, HIGH);
    digitalWrite(greenLED, LOW);
    tone(buzzerPin, 1000);
    Serial.println("FIRE ALERT - PUMP ON");
  } else {
    digitalWrite(pumpPin, LOW);
    digitalWrite(redLED, LOW);
    digitalWrite(greenLED, HIGH);
    noTone(buzzerPin);
    Serial.println("SAFE - PUMP OFF");
  }

  delay(1000);
}
```

RESULT:

The Smart Firefighter IoT System was successfully designed and simulated using Arduino Uno and two TMP36 temperature sensors. The pump-representing motor and buzzer turn ON when either sensor reaches the configured temperature threshold, while the red LED indicates the alert condition. When both sensors are below the threshold, the green LED glows and the pump and buzzer remain OFF.